A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 0.3 kg projectile at 50 m/s sticks to a stationary 2.7 kg cart. Find common speed and the fraction of initial kinetic energy remaining.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × lead time. Toy model: target mass is much larger; straight-line shift comparison. Ejecta can also transfer momentum in a real impact.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Marzano check - circle: ① ② ③

03 Represent and interpret

A tiny sideways velocity change acts for a short time or a long time. Which produces more sideways shift in the straight-line model? Explain.

Marzano check - circle: ① ② ③

04 Solve and justify

Hypothetical impactor: 600 kg at 6000 m/s; target 5×10^9 kg initially at rest in the target frame. Ignore ejecta. EstimateΔv and straight-line shift after 10^6 s.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

A kinetic-impact deflection model

antocciphysics

05 Analyze the reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Marzano check - circle: 1 2 3

06 Choose a method

An asteroid’s crossing window is days 16-20; Earth’s is 18-22. Find the overlap and explain why this is not proof of a collision.

Marzano check - circle: 1 2 3

07 Exit check

A toy target needs a 10 km shift over 2×10^{6} s. Find requiredΔv. If target mass 10^{9} kg, estimate impulse.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 1000 kg hypothetical target needs a transverse velocity change 0.02 m/s. Estimate required impulse. If delivered uniformly for 0.1 s, find average force and displacement from the velocity change after 10^{5} s in a force-free toy model.

Marzano check - circle: 1 2 3